

CONSTRUCTION TECHNOLOGY

Construction technology has always played a central role in society. It is a multifaceted industry in which projects can result in products, systems, processes, or services and encompass various aspects of production, repair, and maintenance. In construction technology courses, students will learn about designing, constructing, and maintaining a variety of buildings and structures, and will gain experience with the tools, equipment, and processes commonly used in the field. Students will also learn about health and safety standards in the construction industry, building codes and regulations, and employment opportunities and careers in the various sectors of the industry, including electricity, carpentry, masonry, heating and cooling, and plumbing.

The construction technology courses initially focus on residential and light construction systems related to residential buildings and structures, and progress to more advanced residential and light commercial construction. The woodworking courses allow students to concentrate on furniture building or cabinet making.

The list of approved emphasis areas for construction technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Custom Woodworking, Grade 11

Workplace Preparation

TWJ3E

This course enables students to develop knowledge and skills related to cabinet making and furniture making. Students will gain practical experience using a variety of the materials, tools, equipment, and joinery techniques associated with custom woodworking. Students will learn to create and interpret technical drawings and will plan, design, and fabricate projects. They will also develop an awareness of environmental and societal issues related to the woodworking industry, and will explore apprenticeships, postsecondary training, and career opportunities in the field that may be pursued directly after graduation.

Prerequisite: None

A. CUSTOM WOODWORKING FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** describe the features and applications of materials and processes used in custom woodworking;
- A2.** describe woodworking tools, equipment, and techniques, and use them safely;
- A3.** use correct terminology related to woodworking and the materials, tools, equipment, and processes involved.

SPECIFIC EXPECTATIONS

A1. Materials and Processes

By the end of this course, students will:

- A1.1** describe traditional and newer natural and manufactured materials commonly used in custom woodworking projects (*e.g., lumber, adhesives, plywood, particle board, oriented-strand board, medium-density fibreboard*);
- A1.2** identify the different classifications and grades that apply to wood used in woodworking (*e.g., hardwood, softwood, clear, select*), and describe the properties and flaws of wood in these classifications (*e.g., knots, shrinkage, warpage*);
- A1.3** describe common woodworking processes (*e.g., material breakout, planing, shaping, sanding, finishing*);
- A1.4** describe the various finishing processes and materials used in custom woodworking (*e.g., processes: painting, staining, veneering, laminating; materials: sealants, stains, varnishes, paints, oils, lacquers, veneers*).

A2. Tools, Equipment, and Techniques

By the end of this course, students will:

- A2.1** describe tools and equipment commonly used in the custom woodworking industry;
- A2.2** demonstrate the ability to use, maintain, adjust, and store woodworking tools and equipment correctly and safely (*e.g., hammers, measuring instruments, saws, drills, lathes*);

- A2.3** describe and competently use common layout, measuring, tracing, and assembly methods (*e.g., layout: determining and marking circumference, radius, and angles; assembly: use of screws, glue, dowels, biscuits, milled joints such as half lap, mortise and tenon, dovetail, and tongue and groove*).

A3. Terminology

By the end of this course, students will:

- A3.1** use correct terminology to identify and describe wood products (*e.g., plywood, veneer, annual rings, kiln-dried lumber, dressed lumber*);
- A3.2** use correct terminology to describe woodworking processes and techniques (*e.g., processes: truing, dressing, joining, levelling, squaring; techniques: shimming, planing, gluing, nailing*);
- A3.3** use correct terminology to identify and describe woodworking tools and equipment (*e.g., air compressor, brad nailer, tri-square, planer, router, T-bevel*);
- A3.4** use correct terminology to describe the elements of woodworking projects (*e.g., frames, panels, joints, drawers, doors, trim, finishes*);
- A3.5** use correct terminology to identify a variety of cabinet and furniture styles (*e.g., cabinet: modular, European, face-frame, custom-built; furniture: Victorian, French Provincial, Scandinavian, art deco, mission, Shaker*).

B. DESIGN, LAYOUT, AND PLANNING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** apply a design process and/or other problem-solving processes and techniques when planning a variety of woodworking projects, taking into account relevant design principles and safety standards;
- B2.** produce and interpret sketches and/or working drawings for a variety of woodworking projects;
- B3.** plan efficient and effective fabrication and assembly of the components of woodworking projects;
- B4.** apply the mathematical skills required in the design and construction of woodworking projects.

SPECIFIC EXPECTATIONS

B1. Design

By the end of this course, students will:

- B1.1** follow the steps of a design process (see pp. 22–23) to plan and develop a variety of woodworking projects (*e.g., kitchen counter, cabinet, entertainment centre, coffee table, desk*);
- B1.2** use appropriate problem-solving processes and techniques (see pp. 21–23) to address challenges related to various woodworking projects;
- B1.3** demonstrate how design concepts (*e.g., structure, aesthetics, ergonomics*) affect aspects of woodworking projects (*e.g., floor plans, prefabrication, choice of materials and finishes*);
- B1.4** use appropriate design elements (*e.g., line, shape, direction, space, texture, colour*) and principles (*e.g., balance, scale, proportion, contrast, unity*) to enhance the appearance and functionality of woodworking projects;
- B1.5** identify the anthropometric measurements and ergonomic considerations for various projects (*e.g., for the height and width of a chair seat or a desk*);
- B1.6** identify and describe the safety standards that apply to woodworking projects (*e.g., Canadian Standards Association [CSA] standards, Underwriters Laboratories of Canada [ULC] standards*).

B2. Working Drawings

By the end of this course, students will:

- B2.1** identify and use appropriate drafting symbols and drawing conventions in drawings of various cabinetry and furniture projects;
- B2.2** produce sketches and/or working drawings (*e.g., orthographic drawings, plans, elevations, details*) by hand and/or with computer-assisted methods, using appropriate metric and/or imperial units;
- B2.3** interpret working drawings accurately to produce woodworking projects (*e.g., determine dimensions, materials, and hardware placement*).

B3. Planning and Preparation

By the end of this course, students will:

- B3.1** identify and plan the interrelationship between structural and decorative elements in woodworking (*e.g., design the structure to support the decorative elements securely, use trim to hide joints and fasteners, choose hardware and finishes that suit the shape and material of a cabinet*);
- B3.2** plan the safe construction and installation of components commonly used in woodworking projects (*e.g., make bills of materials, compile cut lists*);
- B3.3** lay out projects using a variety of tools and equipment (*e.g., tape measure, combination square, trammel points*);

B3.4 describe modifications to a woodworking project that would improve its quality and value (*e.g., using different wood, joints, finishes, or hardware*).

B4. Mathematical Skills

By the end of this course, students will:

B4.1 calculate dimensions for woodworking projects in appropriate metric and/or imperial units;

B4.2 convert between fractions and decimals and between imperial and metric units, using appropriate charts and tables, to determine dimensions and quantities for woodworking projects (*e.g., convert lengths from inches to centimetres, areas from square metres to square feet, and volumes from gallons to litres*);

B4.3 determine lengths and diameters of fastening devices needed to assemble various woodworking projects (*e.g., lengths and gauges of screws, nails, and staples; lengths and diameters of dowels*);

B4.4 apply mathematical concepts and formulas when preparing components of a woodworking project (*e.g., to determine length, circumference, radius, diameter, area, volume, and angles*);

B4.5 prepare estimates of quantities and costs of materials required to complete woodworking projects, using appropriate metric and/or imperial units (*e.g., metres, square metres, litres, board feet, linear feet, square feet*).

C. FABRICATION, ASSEMBLY, AND FINISHING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** fabricate and assemble custom woodworking projects safely, accurately, and efficiently;
- C2.** prepare surfaces and apply finishing products, trim, and hardware correctly and safely.

SPECIFIC EXPECTATIONS

C1. Fabrication and Assembly

By the end of this course, students will:

- C1.1** identify and follow the appropriate procedures to fabricate project components in a logical and efficient sequence (*e.g., select appropriate materials and tools, follow step-by-step instructions*);
- C1.2** use techniques, tools, and equipment to safely and accurately prepare project materials (*e.g., dress raw lumber; measure, cut, square, and drill stock*);
- C1.3** accurately construct projects in accordance with working drawings and specifications;
- C1.4** identify and apply appropriate quality-control measures to ensure precise dimensions and correct assembly (*e.g., accurate measurements and cuts, clean joints, true edges*);
- C1.5** use clamps, fasteners, and adhesives safely and appropriately;
- C1.6** assemble joints safely and accurately (*e.g., butt, lap, mitred, dovetail, and rabbet joints*);
- C1.7** correctly install various components of a cabinetry or furniture project (*e.g., drawer slides, shelf supports, hinges, lazy Susan, casters*).

C2. Finishing

By the end of this course, students will:

- C2.1** prepare surfaces correctly for finish application according to type of material, desired finish, and intended use of the project (*e.g., wood species used, smooth or textured surface, environment in which project will be used*);
- C2.2** select suitable finishes (*e.g., stain, paint, varnish, oil, wax*), taking into account the type of material to be finished, the function of the finish, and the intended use of the project, and use appropriate methods to apply these finishes correctly (*e.g., brush, spray, roller*);
- C2.3** use appropriate techniques, tools, and equipment correctly and safely to install trim and hardware;
- C2.4** select and correctly install appropriate surface materials on counters or furniture (*e.g., laminates, ceramics, butcher block, wood veneer*).

D. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of the environmental effects of the woodworking industry, and ways of reducing harmful effects;
- D2.** describe how the woodworking industry and society affect each other.

SPECIFIC EXPECTATIONS

D1. Technology and the Environment

By the end of this course, students will:

- D1.1** describe the major effects of the woodworking industry on the environment (*e.g., costs and benefits related to forest management; non-sustainable logging that causes deforestation, destruction of old-growth forests, and/or loss of wilderness habitat for endangered species; water and air pollutants released during the production of manufactured materials; energy required to produce and transport materials for woodworking*);
- D1.2** identify the environmental effects of using and disposing of specific natural and manufactured materials (*e.g., landfill or incineration of woodworking waste; harmful emissions from some types of paints, adhesives, and manufactured materials, which contribute to "sick building syndrome"*);
- D1.3** identify ways of reducing environmental harm through the choice of particular materials (*e.g., sustainably produced products, products that have a minimal ecological footprint, non-toxic products*);
- D1.4** plan projects and use materials to minimize waste (*e.g., use efficient cutting patterns, reuse or recycle leftover materials*);
- D1.5** identify various certifications and/or standards for sustainable practices (*e.g., Forest Stewardship Council Canada standards*).

D2. Technology and Society

By the end of this course, students will:

- D2.1** identify the beneficial effects of woodworking for society (*e.g., production of useful and attractive products, enjoyment for hobbyists, therapeutic benefits in senior-citizens' programs*);
- D2.2** describe the economic and cultural effects of the custom woodworking industry (*e.g., creation of jobs within the community, opportunity to preserve and/or apply traditional designs*) on a specific community or population (*e.g., Mennonite, Aboriginal*);
- D2.3** describe how societal needs and client preferences (*e.g., budget restraints; cultural, religious, and environmental choices; ease of use by persons with physical disabilities*) affect custom woodworking projects.

E. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1.** demonstrate an understanding of health and safety regulations and practices specific to woodworking;
- E2.** identify career opportunities in custom woodworking, and describe the training required for these careers.

SPECIFIC EXPECTATIONS

E1. Health and Safety

By the end of this course, students will:

- E1.1** identify hazards related to the materials, processes, and equipment used for custom woodworking (*e.g., flammable or toxic fumes from solvents, paints, and varnishes; risk of injury from cutting tools*);
- E1.2** outline the health and safety legislation, regulations, and standards that apply to the custom woodworking industry (*e.g., Workplace Hazardous Materials Information System [WHMIS], Workplace Safety and Insurance Board [WSIB] regulations, provincial labour legislation, local by-laws*);
- E1.3** describe the rights and responsibilities of employees (*e.g., the right to know, the right to refuse, the right to participate, as outlined in the Occupational Health and Safety Act*);
- E1.4** demonstrate safe practices related to materials, processes, tools, equipment, and facilities used in woodworking;
- E1.5** demonstrate an understanding of when and how to use protective clothing, gear, and equipment (*e.g., ensure proper ventilation and use appropriate protective masks when sanding materials or applying finishes that give off hazardous vapours*).

E2. Career Opportunities

By the end of this course, students will:

- E2.1** identify career opportunities in custom woodworking (*e.g., tradesperson, artisan, technician, technologist*), and describe the education and training required for these careers;

E2.2 explain the importance of lifelong learning for someone choosing a career in the custom woodworking industry;

E2.3 identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the custom woodworking industry (*e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);

E2.4 demonstrate an understanding of and apply the Essential Skills that are important for success in the custom woodworking industry, as identified in the Ontario Skills Passport (*e.g., reading text, document use, measurement and calculation*);

E2.5 demonstrate an understanding of and apply the work habits that are important for success in the custom woodworking industry, as identified in the Ontario Skills Passport (*e.g., reliability, initiative, customer service*);

E2.6 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in custom woodworking (*e.g., Passport to Safety certificate, technical drawings, reports, photographs, Ontario Skills Passport Work Plan*), and explain why having a current portfolio is important for career development and advancement.